

SQL

# Star Schema vs. Snowflake Schema



**Pradeep M**  
@pradeep-m-analyst





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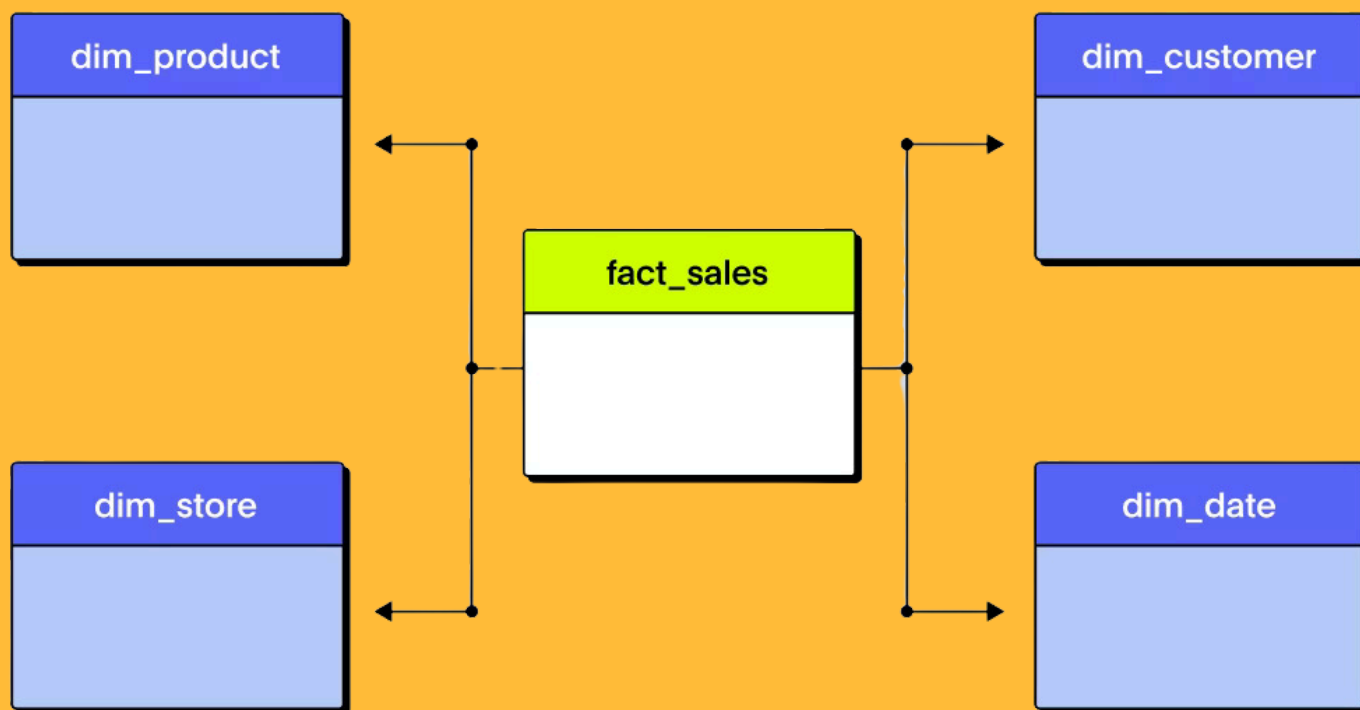
# Star Schema

It is the simplest method for arranging data in a data warehouse.

It contains a **fact table at the center** connected to dimension tables around it.



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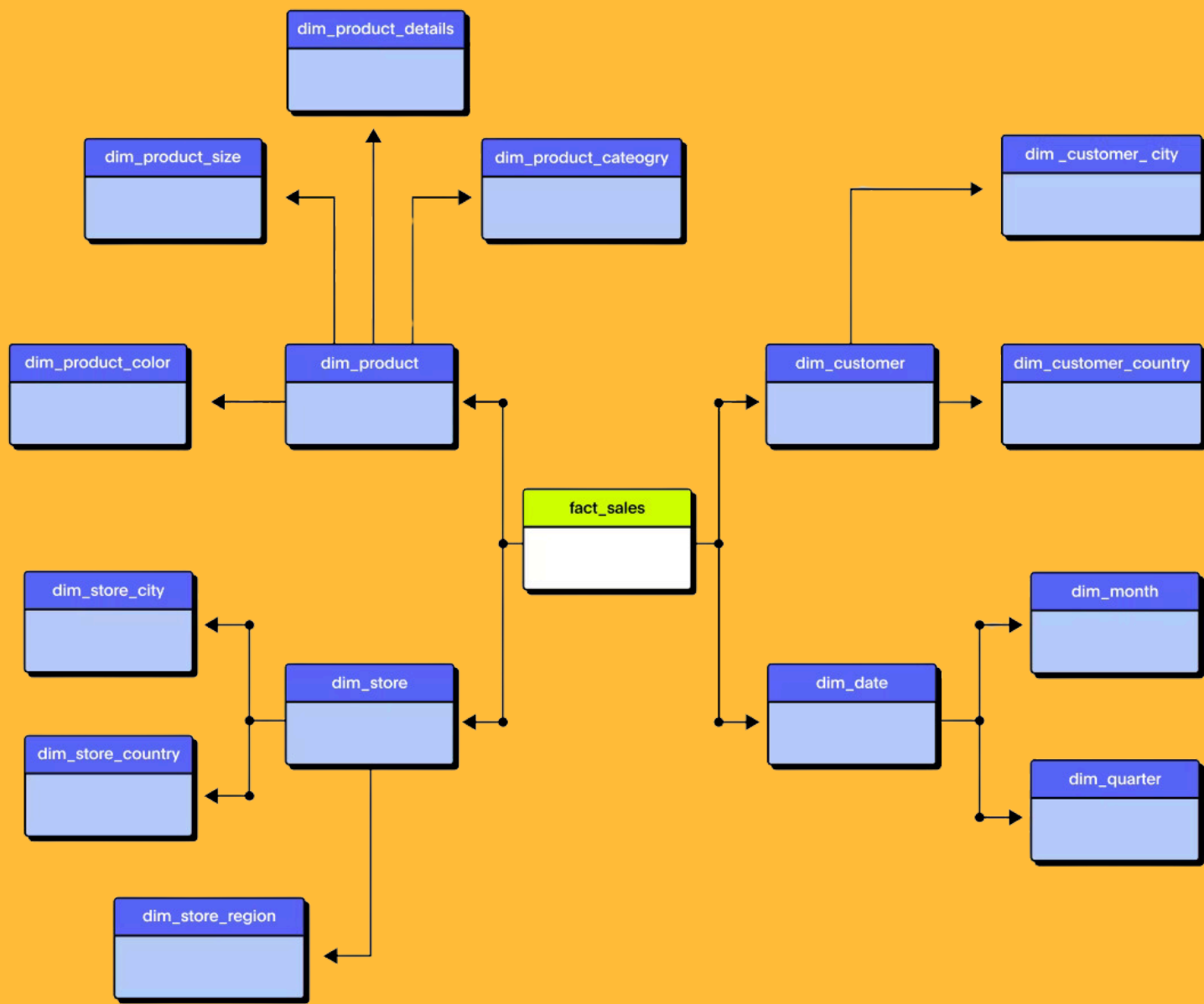


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# Snowflake Schema

It is a more complex method of storing data in which fact tables, dimension tables and sub-dimension tables are connected through foreign keys.

Snowflake is most effective for **in-depth** data query analyses.





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# **Key Differences Between Star Schema and Snowflake Schema**



# Structure

- **Star Schema:** Simple and straightforward with a central table directly connected to surrounding tables.
- **Snowflake Schema:** More complex with a central table connected to multiple levels of related tables.



# Ease of Use

- **Star Schema:** Easier to understand and use because everything is directly connected.
- **Snowflake Schema:** More complicated to use due to multiple levels of connections.





# Performance

- **Star Schema:** Generally faster for querying because of its straightforward structure.
- **Snowflake Schema:** Can be slower for querying because of the additional levels of tables that need to be joined.



# Data Redundancy

- **Star Schema:** May have some data redundancy, but this makes it faster to query.
- **Snowflake Schema:** Reduces redundancy by normalizing data into separate tables, saving storage space.



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# Maintenance

- **Star Schema:** Easier to maintain due to its simple design.
- **Snowflake Schema:** Can be more challenging to maintain because of the complex structure.



# In Summary

- **Star Schema:** Simple, fast, easy to use and maintain, with some data redundancy.
- **Snowflake Schema:** Complex, can be slower, reduces redundancy, and requires more effort to maintain.

**I hope you find this  
valuable! 🥰**

**A like from you would be a great  
encouragement**



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